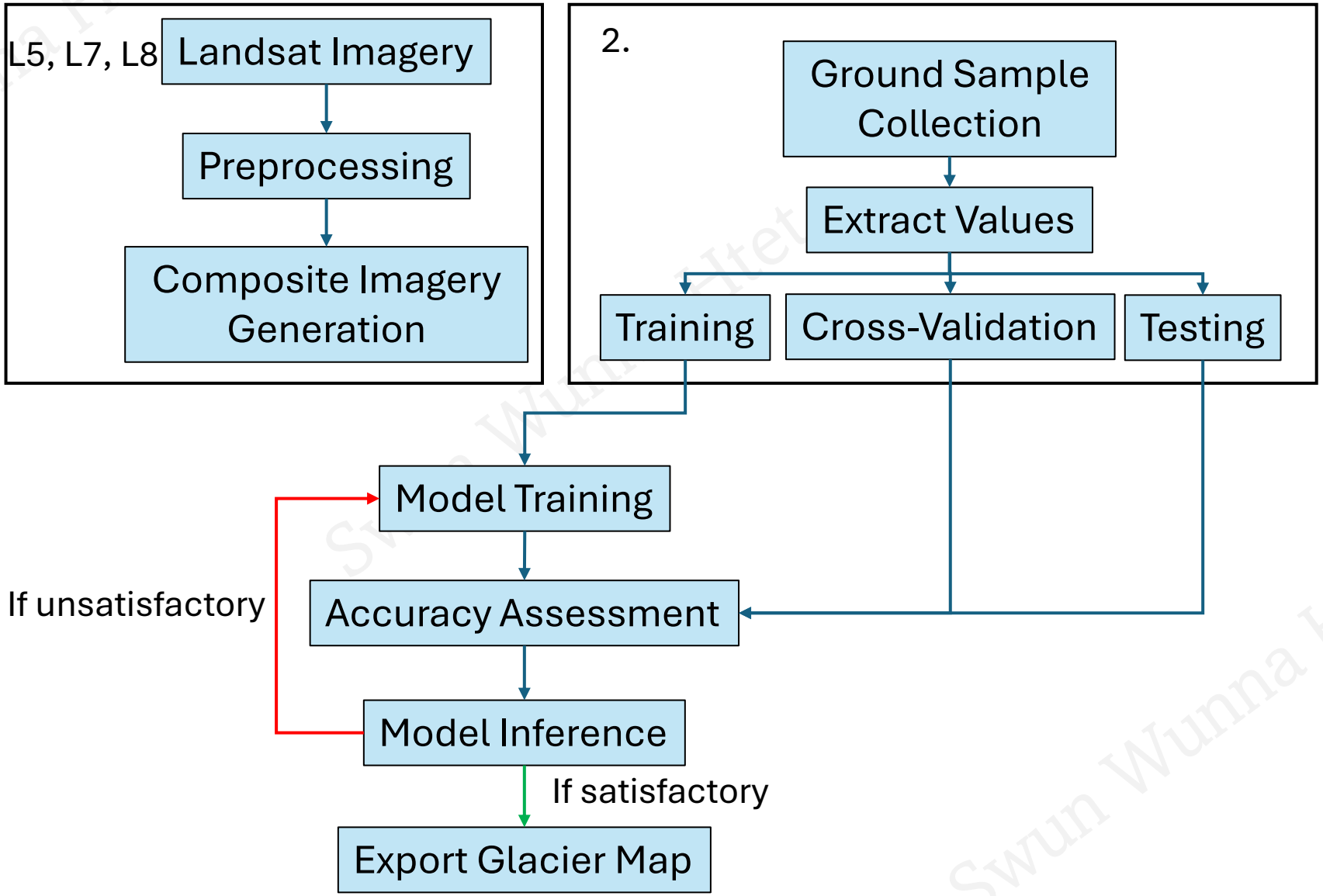
Glacier Mapping Using Google Earth Engine

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Overall Conduct



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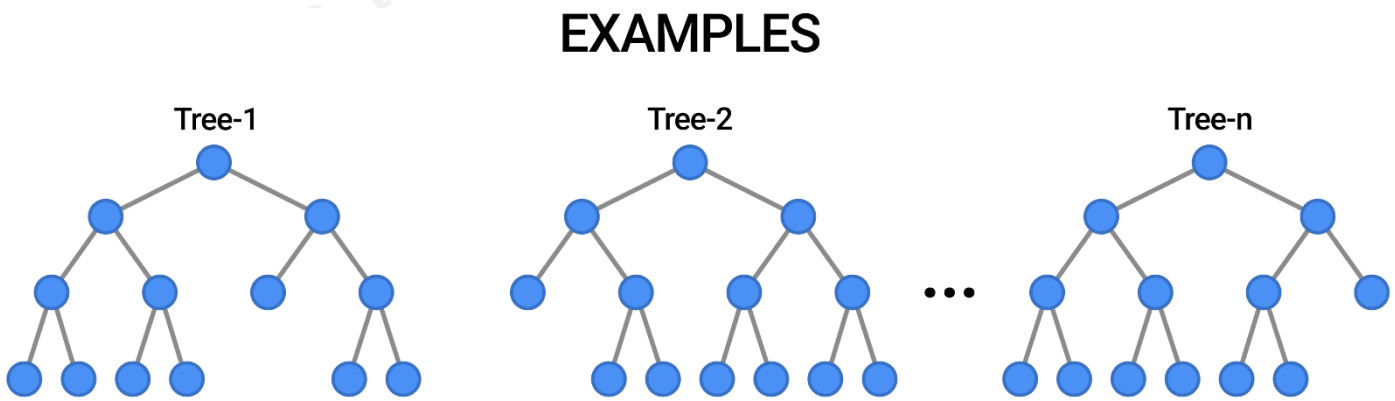
Types of Classification

Supervised Approach	Unsupervised Approach
Training data prepared by users	No pre-determined training dataset
Requires prior user knowledge of input training data property	Using clustering to group pixels related to each other as classes
Random forests, Gradient Boosting, Maximum Likelihood, Support Vector Machine (SVM), Parallelepiped Method, Minimum Distance	K-means Algorithm, ISODATA clustering

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Random Forest

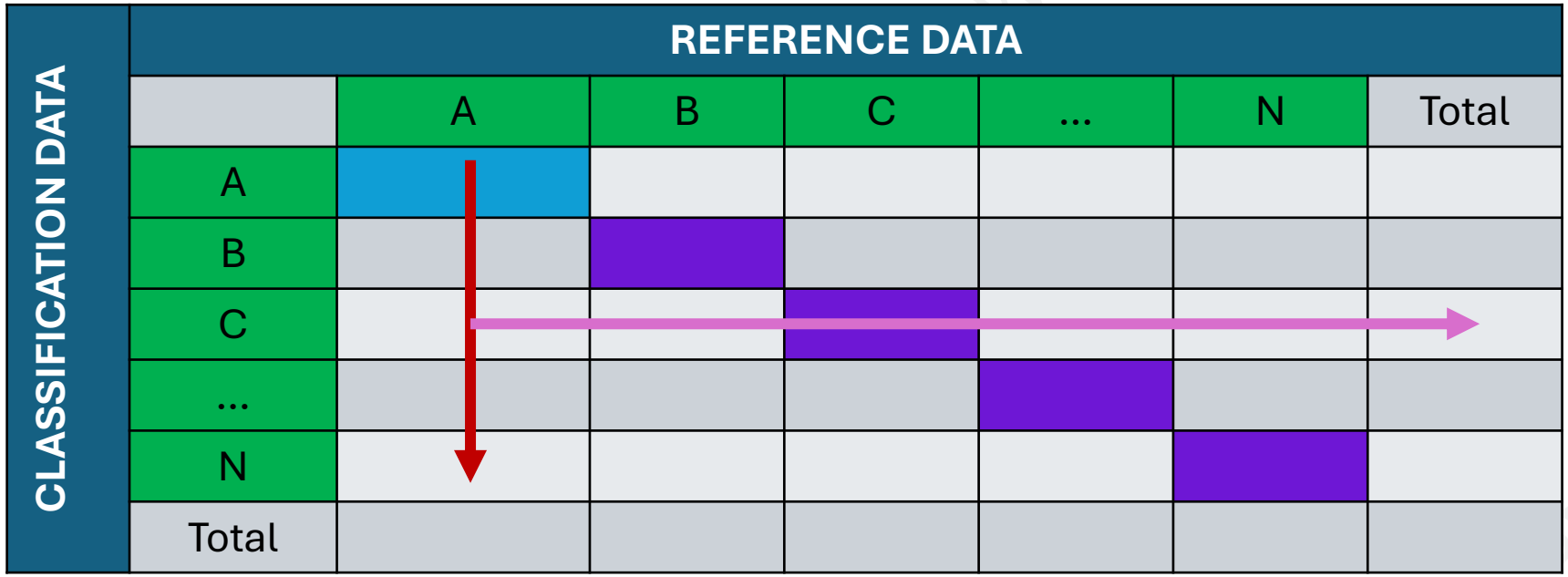
- Ensemble method using decision tree framework.
- **Advantages**:
  - Reducing overfit, unbiased
  - Capable of handling missing data, handle both parametric and non-parametric.
- **Drawbacks**:
  - $Time = cT\sqrt{M}\, N log(N)$
    - c= constant
    - T = Number of trees
    - M = Number of variables
    - N = Number of samples



Confusion Matrix (Error matrix)

- *Error matrix is a comparison table between the reference data and the classified output. Used as the quantitative method of characterizing the image classification accuracy.*

1. $Overall\ accuracy\ (OA) = \frac{Total\ number\ of\ correctly\ classified\ pixels}{Total\ number\ of\ pixels}$
2. User's accuracy (UA) (precision) = probability that a pixel was predicted to be in a certain class is that class
3. Producer's accuracy (PA) (recall) = probability that a pixel in a class was classified correctly.
4. Commission error (EC) = proportion of a pixel that was predicted to be in a class but do not belong to that class
5. Omission error (EO) = proportion of the observed pixel on the ground that are not classified on the map



Column totals = total number of reference pixels in each class

Raw totals = total number of classified pixels in each class

Diagonal totals = total number of pixels correctly classified according to the reference data

Source: Congalton, R. G. (2001). Accuracy assessment and validation of remotely sensed and other spatial information. *International Journal of Wildland Fire*, *10*(3–4), 321–328. https://doi.org/10.1071/wf01031

Example

CLASSIFICATION DATA	REFERENCE DATA				
	Classes	A	B	C	Row Total
	A	24	10	4	38
	B	5	30	2	37
	C	1	1	23	25
	Column Total	30	41	29	100

$$OA = \frac{Diagonal\ total}{Total\ pixels} = \frac{24 + 30 + 23}{100} = 77\%$$

$$PA_A = \frac{Diagonal\ value\ (A)}{Total\ pixels\ in\ row\ (A)} = \frac{24}{38} = 63\%$$

$$PA_B = \frac{Diagonal\ value\ (B)}{Total\ pixels\ in\ row\ (B)} = \frac{30}{37} = 81\%$$

$$PA_C = \frac{Diagonal\ value\ (C)}{Total\ pixels\ in\ row\ (C)} = \frac{23}{25} = 92\%$$

$$UA_A = \frac{Diagonal\ value\ (A)}{Total\ pixels\ in\ column\ (A)} = \frac{24}{30} = 80\%$$

$$UA_B = \frac{Diagonal\ value\ (B)}{Total\ pixels\ in\ column\ (B)} = \frac{30}{37} = 81\%$$

$$PA_C = \frac{Diagonal\ value\ (C)}{Total\ pixels\ in\ row\ (C)} = \frac{23}{25} = 92\%$$

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Source: Cindy Schmidt & Amber McCullum: *Assessing the Accuracy of Land Cover Classifications (ARSET Training) (Feb 13, 2018)*

Kappa Statistics

- The kappa coefficient is another accuracy indicator can be estimated easily from the confusion matrix or error matrix that is widely used in classification accuracy assessment.
- Value ranges from 0 (inaccurate) to 1 (perfect).

Kappa Coefficient	Classification can be regarded as
Below 0.4	Poor
0.41 - 0.60	Moderate
0.61 - 0.75	Good
0.76 - 0.80	Excellent
0.81 and above	Almost Perfect

$$k = \frac{Observed\ - Expected}{1\ - Observed}$$

Where: k = Kappa coefficient (0 – 1)

Observed = probability of correct classification (Observed Accuracy)

Expected = probability of change agreement

F1-Score

- Combining precision and recall into one harmonic mean representative value.

$$F1 = 2\ \times \frac{Precision\ \times Recall}{Precision\ \times Recall}$$

F1-Score	Interpretation
> 0.9	Very good
0.8 - 0.9	Good
0.5 – 0.8	Ok
<0.5	Not good

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Landsat Image Collection Filtering

- As usual, we will select our landsat image collection.

```
////////////////////////////////////////////////////////////////////////
// DEFINE YOUR SET ANALYSIS PERIOD
////////////////////////////////////////////////////////////////////////
var YEAR = 1998;
var start = ee.Date(YEAR+'-08-01');
var end = ee.Date(YEAR+'-10-01');

var satellite = 'L5'; // (Mandatory) L5 for 1993, L8 for 2025

////////////////////////////////////////////////////////////////////////
// FILTER DATASETS
////////////////////////////////////////////////////////////////////////


// For 1993, use Landsat 5, for 2025 use Landsat 8
if (satellite == 'L5'){
  var col = l5.filterBounds(aoi).filterDate(start, end);
}else{
  var col = l8.filterBounds(aoi).filterDate(start, end);
}
```

Note that we select August-September date range (obtained from ablation period determination)

Conditional selection of landsat collection

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Quality Masking

```
////////////////////////////////////////////////////////////////////////
// FUNCTION FOR QUALITY MASKING FOR LANDSAT IMAGERY (BOTH L5 AND L8)
////////////////////////////////////////////////////////////////////////

function QAMASK(img){
  // Preprocessing all the ee.Images available in country boundary
  //
  var qa = img.select('QA_PIXEL');

  // selecting the mask bits
  var dil_cloud = qa.rightShift(1).bitwiseAnd(1);
  var cirrus = qa.rightShift(2).bitwiseAnd(1);
  var cloud = qa.rightShift(3).bitwiseAnd(1);
  var cloudshadow = qa.rightShift(4).bitwiseAnd(1);

  // keeping the valid mask
  var mask = dil_cloud.eq(0).and(cirrus.eq(0)).and(cloud.eq(0)).and(cloudshadow.eq(0));

  var output = img.updateMask(mask).copyProperties(img, ['system:time_start', 'WRS_PATHROW']);

  return ee.Image(output);
}
```

QA_PIXEL Bitmask

- Bit 0: Fill
- Bit 1: Dilated Cloud
- Bit 2: Cirrus (high confidence)
- Bit 3: Cloud
- Bit 4: Cloud Shadow

1 (Yes)/ 0 (No)

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All-in-one Preprocessing

```
////////////////////////////////////////////////////////////////////////////////
// FUNCTION FOR RADIOMETRIC CONVERSION, NDSI AND TEMPORAL REDUCTION
////////////////////////////////////////////////////////////////////////////////
function PREPROCESS(col, operation){

  var revised = col.map(function(img){
    var img2 = img.select('SR_B.*');

    img2 = (img2.multiply(0.0000275)).subtract(0.2);

    // img2 = img2.focalMean(1, 'square','pixels', 3);

    var ndsi = img2.normalizedDifference(['SR_B2', 'SR_B5']).rename('NDSI');
    img2 = img2.addBands(ndsi);

    // Conversion for thermal Band
    var img3 = img.select('ST_B6');
    img3 = (img3.multiply(0.00341802)).add(149).toFloat().rename('ST_B6');
    img2 = img2.addBands(img3);

    return ee.Image(img2.copyProperties(img, ['system:time_start', 'WRS_PATHROW']));
  });

  var median;
  if (operation == 'min'){
    median = revised.min().clip(aoi);
  }else if (operation == 'max'){
    median = revised.max().clip(aoi);
  }else{
    median = revised.mean().clip(aoi);
  }

  return ee.Image(median.copyProperties(revised, ['system:time_start', 'WRS_PATHROW']));
}
```

Radiometric Conversion

Calculate NDSI band

Radiometric Conversion of LST band

Apply reducer operation based on argument
‘min’ = minimum
‘max’ = maximum
‘mean’ = mean

```
////////////////////////////////////////////////////////////////////////////////
// NOW, WE'RE PREPROCESSING THE LANDSAT COLLECTION
////////////////////////////////////////////////////////////////////////////////

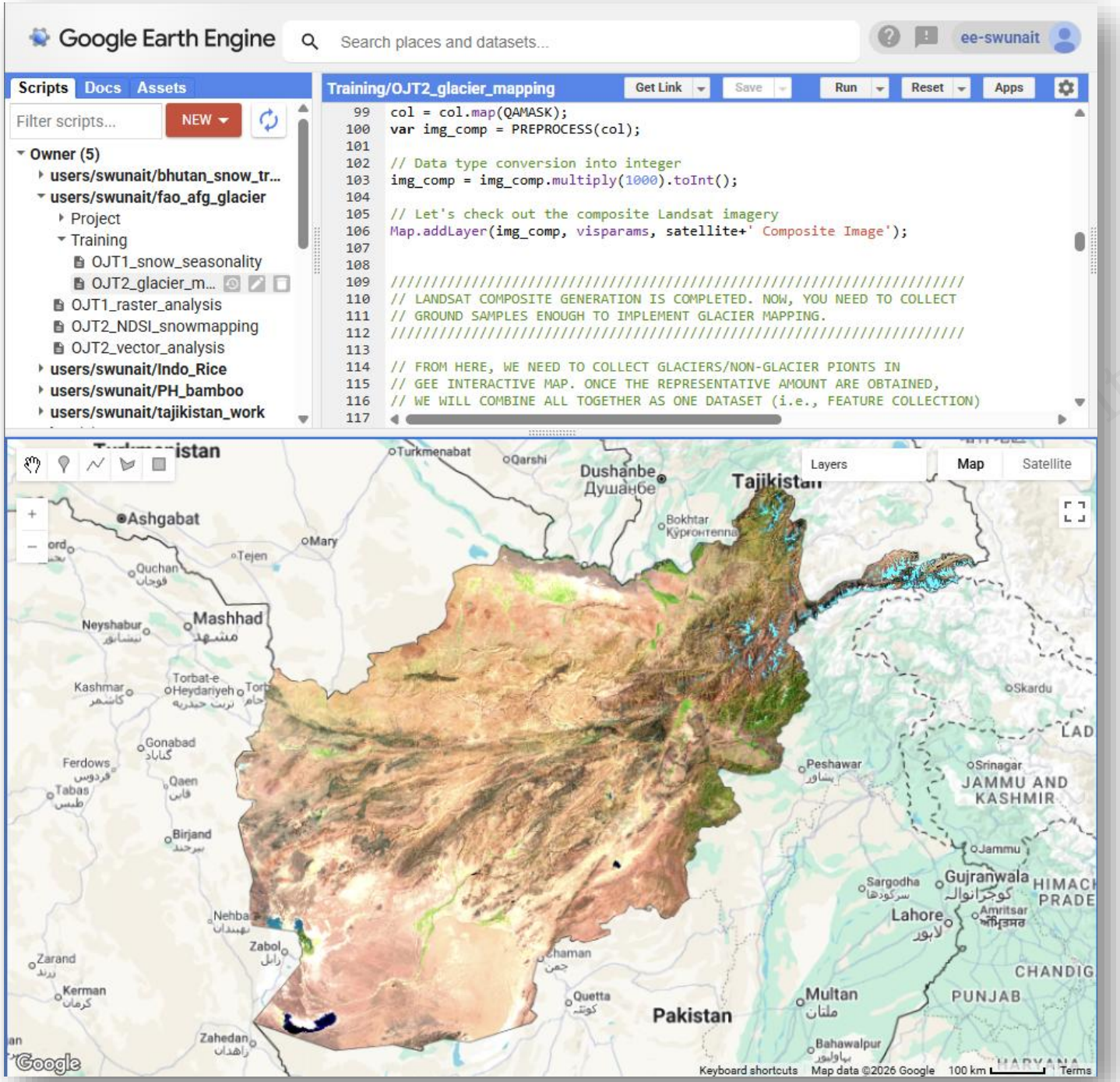
col = col.map(QAMASK);
var img_comp = PREPROCESS(col);

// Data type conversion into integer
img_comp = img_comp.multiply(1000).toInt();

// Let's check out the composite Landsat imagery
Map.addLayer(img_comp, visparams, satellite+' Composite Image');
```

In short, we use all multispectral bands, LST, NDSI and input features for glacier mapping.

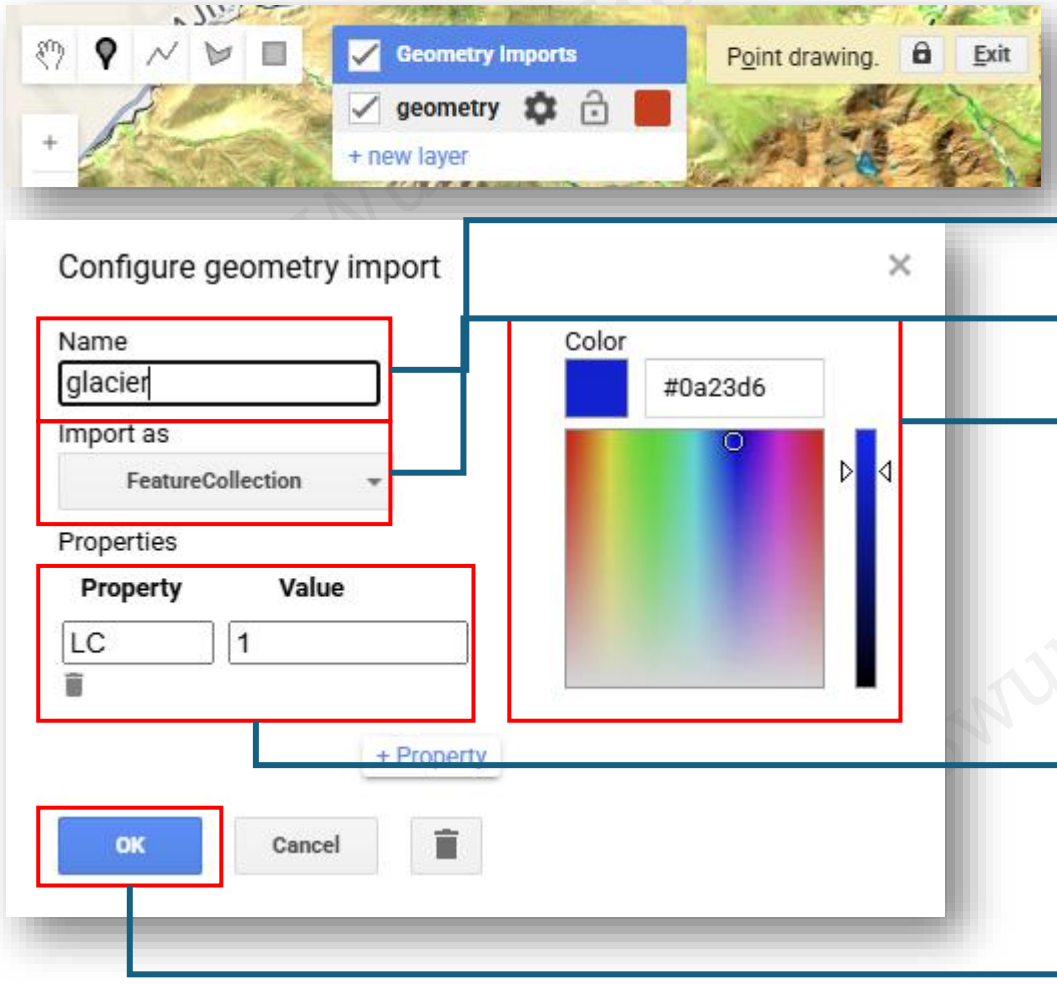
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Landsat Composite Imagery for 2025

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Now, we can start off with ground sample collection



Name the geometry as "Glacier"

Select as Feature Collection

Select a color

Provide the attribute
LC = Land Cover
Value = 1 (Glacier)

Click OK

- Try to collection both glacier and non-glaciers to nearly equal sample sizes.
- For non-glaciers, sampling areas outside of glacier area is highly recommended.

Repeat the same routine for Non-glacier Feature Collection
Name : Other, LC = 0 (Non-glacier)

Merge both datasets and starts extracting multispectral information

```
// FROM HERE, WE NEED TO COLLECT GLACIERS/NON-GLACIER PIONTS IN
// GEE INTERACTIVE MAP. ONCE THE REPRESENTATIVE AMOUNT ARE OBTAINED,
// WE WILL COMBINE ALL TOGETHER AS ONE DATASET (i.e., FEATURE COLLECTION)
// LABEL INFORMATION: 1 = GLACIER, 0 = OTHERS (NON-GLACIER)

var training = glacier.merge(other);

// OR, IF YOU ALREADY HAVE THE LOCATION POINTS UPLOADED IN THE GEE ASSETS.
// NOTE THAT YOU NEED AN ATTRIBUTE COLUMN 'LC' TO CLASSIFY COLUMNS


// var training = ee.FeatureCollection('PROVIDE HERE');

//////////////////////////////////////////////////////////////////
//    GLACIER CLASSIFICATION STARTS HERE
//////////////////////////////////////////////////////////////////

// NEXT, WE'RE GOING TO EXTRACT THE VALUE FROM THE REFERENCE IMAGE
var sampled = img_comp.sampleRegions({
  collection: training,
  properties: ['LC'],// Provide your column name.
  scale: 30,
  geometries: true
});
```

Merged sample points as feature collection

Extracting multispectral information

Notice that LC is point's attribute
Scale is set to 30 meters.

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In case you're interested for Exploratory Data Analysis

```
// EXPORT THE EXTRACTED MULTISPECTRAL VALUES AS SHAPEFILE TO ASSESS
// EXPLORATORY DATA ANALYSIS; Comment out the rest of the code for EDA
// to execute separately.

Export.table.toDrive({
  collection: sampled,
  description: YEAR+'_training_samples',
  folder: 'FAO Afghanistan Glaciers Project', // Provide your google drive folder
  fileFormat: 'SHP',
  selectors: ['LC', 'NDSI', 'SR_B1', 'SR_B2', 'SR_B3',
              'SR_B4', 'SR_B5', 'ST_B6',
              'SR_B7']
});
```

- Ground sample feature collection with extracted multispectral information can be exported as shapefile for archive purposes and as support for exploratory data analysis (EDA).

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Splitting the ground truth dataset into three categories

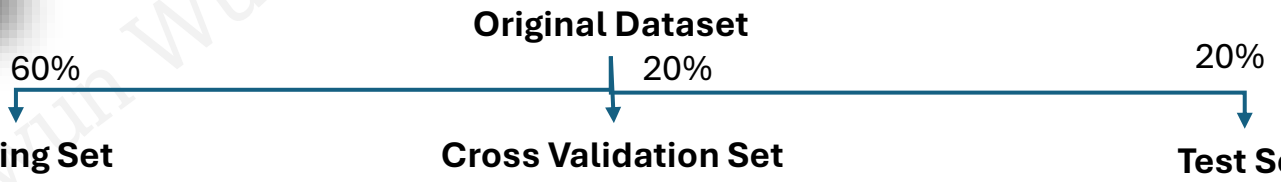
```
// We'll split the whole dataset into three parts:
// 1. Training set (60%)
// 2. Cross-validation set (20%)
// 3. Testing set (20%)

sampled = sampled.randomColumn('random');
// print("Dataset Columns: ", sampled.limit(10));

var train_set = sampled.filter(ee.Filter.lt('random', 0.6));

var cv_set = sampled.filter(
  ee.Filter.and(
    ee.Filter.gte('random', 0.6),
    ee.Filter.lt('random', 0.8)
  )
);
var test_set = sampled.filter(ee.Filter.gte('random', 0.8));
```

Creating a column of random value.
After that, randomly select 60 % of the data as training set.
After that, randomly select 20% of the data as cv set.
The remaining 20% would be the testing set.



Training Set

		Reference	
		Glacier	Other
Prediction	Glacier	97	0
	Other	0	91
F1-Score	Glacier	1.	
	Other	1.	
Overall Accuray		1.	

Cross Validation Set

		Reference	
		Glacier	Other
Prediction	Glacier	30	0
	Other	0	17
F1-Score	Glacier	1.	
	Other	1.	
Overall Accuray		1.	

Test Set

		Reference	
		Glacier	Other
Prediction	Glacier	38	0
	Other	0	29
F1-Score	Glacier	1.	
	Other	1.	
Overall Accuray		1.	

Using training set, train the random forest model

```
// NOTE THAT THE MODEL IS TRAINED USING TRAINING SET, NOT WITH THE WHOLE DATASET.
var RF_classifier = ee.Classifier.smileRandomForest({
  numberOfTrees: 100,
  variablesPerSplit: 2,
  minLeafPopulation: 1
}).train({
  features: train_set, // PAY ATTENTION TO THIS
  classProperty: 'LC',
  inputProperties: img_comp.bandNames()
});
```

Model setting for training

Note that we're using training set to train the model. Not otherwise.

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After model training, model inference can be applied.

```
//////////////////////////////////////////////////////////////////////
// WHEN THE ACCURACY ASSESSMENT IS DONE, WE WILL RUN CLASSIFICATION
//////////////////////////////////////////////////////////////////////

var classified_map = img_comp.classify(RF_classifier).toByte();

// Visualizing
Map.addLayer(classified_map, {}, 'Glaciers');


//////////////////////////////////////////////////////////////////////
// FINALLY, WE'RE EXPORTING CLASSIFIED MAP IN GOOGLE DRIVE
//////////////////////////////////////////////////////////////////////
Export.image.toDrive({
  image: classified_map,
  description: 'AFG_glacier_map_'+YEAR,
  folder: 'FAO Afghanistan Glaciers Project',// CHANGE YOUR GOOGLE DRIVE FOLDER HERE
  region: aoi,
  scale: 30,
  crs: 'EPSG:4326',
  maxPixels:1e12
});

Export.image.toDrive({
  image: img_comp,
  description: 'AFG_'+YEAR+'_Imagery',
  folder: 'FAO Afghanistan Glaciers Project', // CHANGE YOUR GOOGLE DRIVE FOLDER HERE
  region: aoi,
  scale: 30,
  crs: 'EPSG:4326',
  maxPixels:1e12
});
```

Apply classification to the created landsat composite image using trained model.

Exporting the classified map and Landsat imagery

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Check out the obtained glacier map with Landsat imagery

